

# Physical Action Effects in Dexterous Interaction Models: Measurement, Randomized Identification, and Task-Utility Prerequisites

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## Abstract

We study whether an interaction model can distinguish alternative physical commands before investing in dexterous policy optimization. Initial wrist-pulse, contrast-learning and actuation-factorization screens fail their fixed gates. Geometry-conditioned cold replay acquires all 384 windows but fails its strict pre-intervention matching requirement. Prospective randomized assignment instead identifies conditional-effect risk differences without claiming solver-state clones. On 3072 new windows, archived nominal-motion predictors show no robust advantage over command or zero-effect controls. A directly contrast-supervised learner, trained on these randomized windows with crossfit nuisance centering, also fails on 3072 further windows: direct-minus-factual risk difference 23.98 mm squared. Exploratory comparisons between its frozen controls suggest a simpler factual learner merits a separate task test. All failures remain explicit. These results establish measurement and route-selection evidence on one simulation object; no new general modeling method, manipulation-policy benefit or journal readiness is claimed.

## 1. Problem and relation to prior work

An interaction model can fit common motion while failing to compare alternative actions. Useful comparisons additionally require experiments capable of measuring physical differences at the states in which actions are selected. We examine this prerequisite before committing further resources to a policy. The repository provides self-trained actors, geometric models and a DExplore simulation substrate [xu2025]. Historical policy audits are motivation rather than fresh evidence for this manuscript.

Shared particle actions and cross-embodiment world models already support planning [he2025]. Value-aware objectives [farahmand2017], predictor-level action recovery [qiu2026] and multi-step controllability representations [rudolph2024] are established. Contact-consistent replay and actuator residuals are also developed by Contact-UAN [fey2026]. Our small physical MLP studies do not reproduce these systems or establish priority for their general principles. The present contribution is an exploratory evidence record with frozen hypotheses and retained failures.

## 2. Initial physical and contrast-learning screens

Four panels combine actor seeds 286/287 with environment seeds 288/289, each with 32 environments for each of three airplane motions. A retrospective trigger schedules one step after the first reference event where independent hand and object force proxies are active. A one-step wrist-z action change requests a plus or minus 10-mm PD target offset. This is a desired offset, not measured wrist travel. All subsequent normalized actions and RNG schedules are replayed. Two zero arms measure repeated process variation. Each object response subtracts its own actual pre-intervention position. Pulse and repeat contrasts are summarized by vector RMS.

| Steps | Pulse RMS (mm) | Repeat RMS (mm) |
|-------|----------------|-----------------|
| 1     | 1.914          | 0.000           |
| 2     | 2.572          | 0.084           |
| 5     | 7.150          | 0.101           |
| 10    | 18.827         | 0.444           |
| 20    | 75.463         | 6.081           |

| Steps | Pulse RMS (mm) | Repeat RMS (mm) |
|-------|----------------|-----------------|
| 30    | 105.792        | 9.638           |

Table1. Initial fixed panels: pulse versus observed zero-repeat position contrasts.

At five and ten steps, contrast ratios are 70.44 and 42.36. At one step, the observed zero repeat is exactly zero; its ratio is undefined rather than made finite by an arbitrary denominator. Mixed vertical signs fail the joint physical gate. No beneficial lifting or population confidence bound follows from a large observed contrast.

The subsequent vector-learning decision retains this failure. It fits 192 actor-286 windows and tests 192 actor-287 windows using four past/current physical states and ten preset replay actions. Future observed physical states are absent from inputs. Matched two-layer 128-unit SiLU networks predict five/ten-step position responses; three seeds each receive 1000 Adam updates. Differential training adds measured plus-minus contrast loss. All normalization uses fit rows. Fixed test panels were already inspected by the physical experiment, making this reused-data exploration.

| Input/objective              | Five steps (mm) | Ten steps (mm) |
|------------------------------|-----------------|----------------|
| Factual                      | 10.121          | 26.546         |
| Differential                 | 10.114          | 26.990         |
| No pulse                     | 9.922           | 26.456         |
| Global fit mean              | 10.061          | 26.259         |
| Motion fit mean (privileged) | 10.137          | 26.340         |

Table2. Actor-transfer contrast RMSE, millimeters; learned rows average three optimization seeds.

Differential training does not improve over factual or global fit-mean controls. Vertical sign agreement is 32.6%/31.1%, below its fixed 75% gate. Both initial screens are UNPROMISING. Increasing steps or choosing another seed is not used to salvage these exact designs. Physical acquisition took 481.9 seconds; fitting took 17.2 seconds on one RTX 3090.

### 3. Geometry and larger archived predictive screen

Full sampled current geometry places 318 of 384 old triggers within 20 mm. However, actor 287's motion 1 is entirely outside that range in both old test panels, with gaps 55-105 mm. Independent force proxies can both activate due to table contacts. The geometric shift is documented, without identifying it as the sole cause of learning failure or redefining the old gate.

We next use two immutable, episode-disjoint historical collections, 960 complete episodes each. Nonterminal frames at step modulo 16 equal to zero yield 32902 fit and 32896 test rows. These collections have been used by previous research and are not pristine Validation data. The current-near test stratum contains 1756 frames from 639 episodes; proximity uses a frozen stride-8 surface convention.

The target is one-step object-position residual over constant velocity, expressed in the current object frame. State input includes joint angles and velocities, object velocities, gravity direction, object height, palm/tip relative positions and force proxies. Matched effect networks receive either zero action slots, native commands, exact PD-target/FK palm-tip motion, predicted realized hand motion, or measured future hand motion. The last is a privileged diagnostic. An independently fit actuator model predicts joint residuals before the learned-motion variant's FK. Effect networks have two 128-unit SiLU layers, fixed 1000 updates and three seeds. Sampling is half-near and half-all fit rows. No test-selected checkpoint is used.

| Effect input                      | Near RMSE (mm) | All RMSE (mm) |
|-----------------------------------|----------------|---------------|
| State only                        | 17.180         | 7.328         |
| Native command                    | 16.436         | 7.330         |
| Nominal PD motion                 | 14.599         | 7.093         |
| Predicted hand motion             | 15.066         | 7.206         |
| Measured future hand (privileged) | 13.994         | 7.001         |
| Passive                           | 23.400         | 7.587         |

Table3. Archived one-step vector prediction error, seed-mean millimeters. Near and all-state strata differ.

The predicted-motion primary method gains 8.33% over command input, failing the 10% requirement despite beating state-only, passive and its geometric transport control. Its gate remains UNPROMISING. Actuator wrist-position errors are 11.75-12.07 mm near the object. Measured future hand motion includes possible contact feedback and cannot establish a deployable causal mediator.

The separately reported nominal-motion variant gains 11.18% and is better in each seed. This motivates a separately frozen fresh causal test; it does not retroactively replace the primary method. The label-free nominal no-slip transport control is inaccurate, with near RMSE 133.64 mm. The full screen took 53.64 seconds. All 30 saved held RMSE metrics were independently recomputed.

## 4. Fresh proximity acquisition and failed matching requirement

New environment seeds 490/491 are combined with the same two actors. A compact reference records up to 360 steps, stopping at the first native termination. For each environment, acquisition selects the first pre-action step at or after step 10 with five remaining nonterminal transitions, both current force proxies, and full sampled surface distance at most 20 mm. Future object outcomes never choose a trigger. Frozen assets provide 1538 hand and 1024 object points.

All 384 environments acquire a qualifying state, including all three motions in every panel. The maximum sampled gap per panel is 16.60, 15.57, 18.83, 18.12 mm. Eight cold processes per panel implement two zeros and one-step plus/minus pulses in world x, y, z. Initial tensors, physical properties, actor/RMS states, actions and RNG schedules are checked. Warm solver state is not asserted to be cloneable. The existing effect models are frozen before acquisition; no refit is allowed.

The intended comparison requires pre-intervention position RMS at most 0.05 mm, maximum joint difference at most 0.0001, and maximum quaternion-component difference at most 0.0001. Joint coordinates combine translations in meters and angles in radians; quaternion differences are dimensionless component differences.

| Panel     | Position RMS (mm) | Joint max | Quaternion max |
|-----------|-------------------|-----------|----------------|
| t286_s490 | 0.003554          | 0.0000732 | 0.0000670      |
| t286_s491 | 0.010859          | 0.0002204 | 0.0006188      |
| t287_s490 | 0.007110          | 0.0000000 | 0.0000416      |
| t287_s491 | 0.013398          | 0.0000982 | 0.0000881      |

Table4. Maximum pre-intervention mismatch over arms relative to zero-a. Limits: position 0.05 mm; joint and quaternion components 0.0001.

All 32 physical arms complete, but seven arms of panel actor 286/seed 491 fail the joint/orientation precondition. The first detected failure is zero-b, before any action intervention. Execution stops with FAILED; the scientific transfer screen is UNCLEAR due to invalid matching. No favorable arm or window is retained, no threshold is widened, and no

pooled causal score or model winner is reported. Acquisition and the failed analysis took 1152.21 seconds. This is a measurement blocker, not a refutation of the nominal predictor.

## 5. Exact-null recovery mechanism screen

Six native commands are overwritten by dependent finger targets. Randomizing them leaves every PD target exactly unchanged; all 32896 held rows verify this bitwise. We test whether recovering such independent null commands from model-generated object predictions induces false action responses. A truthful physical transition cannot recover an independent normalized null coordinate with population MSE below one. This is an elementary conditional-variance bound, not a novel theorem. Mutual-information sufficiency limits are already studied [rakelly2021].

Every minibatch independently randomizes null commands while keeping physical labels fixed. Four matched physical predictors receive factual loss only, full inverse loss, effective-command inverse loss, or hard-canonicalized forward input with effective inverse loss. Inverse heads take current state and generated object residuals, never actual future state. Seeds 511-513 each train for 1000 updates; the auxiliary coefficient is fixed at one. Hard canonicalization guarantees null invariance by construction, without establishing learned causal correctness.

| Variant                 | Near RMSE (mm) | Null response (mm) | Null inverse MSE |
|-------------------------|----------------|--------------------|------------------|
| Factual only            | 16.557         | 1.118              | not trained      |
| Full inverse            | 16.534         | 1.079              | 1.002            |
| Effective inverse       | 16.531         | 1.069              | 1.023            |
| Hard quotient + inverse | 16.564         | 0.000              | 1.023            |

Table5. Exact-null intervention diagnostics, near stratum and seed means. Untrained null-head coordinates in effective/quotient variants are diagnostic only.

Full inverse null error remains near the population floor and its null response is below factual-only in every seed. Three of five interference gates fail: UNPROMISING. The proposed inverse-induced null-encoding mechanism is not supported here. Hard canonicalization removes null response, with little factual-error change. This screen took 33.06 seconds; 66 saved prediction/recovery metrics were independently audited. It is neither an AD-WM reproduction nor a general rejection of inverse learning.

## 6. Prospective randomized identification

To avoid invalid cold-replay matching, we acquire a single factual transition per independently assigned intervention. Four 768-environment panels combine actors 286/287 with initial seeds 492/493, 256 environments per motion. Assignment is drawn with a private CPU generator before physics. The trigger is the first even current step at or after 10 with both force proxies and full sampled hand-object gap at most 20 mm. A zero arm or one-step plus/minus wrist pulse in world x, y, z is executed once; the actor remains closed loop. All 3072 windows complete. Realized counts per motion are 37/37/37/37/36/36/36; recorded known propensities account for the unequal count convention.

Let  $Y$  be observed one-step world position residual over current object velocity. For each wrist axis  $j$ , define  $Z_j$  as the observed residual times the difference between the inverse-propensity indicators for its plus and minus arms. Under randomization and the independent-environment potential-outcome assumption, its conditional mean is the plus-minus physical effect  $\tau_j(X)$ . For predictions  $f, g$ , the row score is  $\sum(f^2 - g^2 - 2(f-g)Z)$ , summing all three output coordinates and intervention axes. Its expectation is the squared conditional-effect risk of  $f$  minus that of  $g$ . This elementary estimator identifies a difference, not absolute effect RMSE or individually observed counterfactuals. Weak numeric interactions between batched environments remain an assumption limitation; exact solver-state cloning is not asserted.

Existing nominal-motion and command checkpoints 411-413 are frozen before the trial. Each scores all six candidate commands at the same current context. Per-model losses are averaged across seeds. Descriptive intervals use 2000

bootstrap replicates, preserving shared initial-seed/environment blocks across actors within six initial-seed/motion strata.

| Comparison            | Point (mm <sup>2</sup> ) | Central95%     | Upper95% |
|-----------------------|--------------------------|----------------|----------|
| nominal minus command | 0.122                    | [-0.919,1.187] | 1.013    |
| nominal minus zero    | -0.326                   | [-1.985,1.350] | 1.124    |
| command minus zero    | -0.448                   | [-1.393,0.455] | 0.296    |

Table6. Randomized held-window conditional-effect risk differences, mm squared; negative favors the first predictor.

All three nominal-motion gates fail: UNPROMISING. Its historical 11.18 percent factual error advantage does not establish causal action-effect superiority. The trial takes 137.84 seconds after a retained 35.85-second device-alias engineering failure. All 3072 physical windows, risk statistics and 107 input hashes pass an independent numeric audit. We stop direct control use of these archived models.

## 7. Direct randomized learning and strong controls

The 3072 observed randomized windows now become fit data. A new test uses initial seeds 494/495 and all 3072 windows; no future test outcome chooses a model. This cohort changes the assignment protocol explicitly to independent seven-arm draws with propensity 1/7, avoiding forced assignment balance. Counts are recorded as realized; no rebalancing or partial-window selection is performed.

Context comprises 63 current physical/geometry features and 18 reference commands. Six overwritten finger-command inputs are canonicalized to zero. Motion identity and observed future states are excluded. Two nuisance folds separate acquisition seeds 492/493, keeping shared actor/environment blocks together. Each nuisance model predicts Y from context, scales only its own fit fold, and predicts the other fold. Subtracting this out-of-fold response mean before constructing the inverse-propensity pseudo-contrast preserves its conditional mean. Crossfitting and randomization are established methods, not novelty claims.

For seeds 611-613, direct models have 81 inputs, 128/128 SiLU layers and nine outputs reshaped as a world contrast matrix. Factual controls receive the same context plus six pulse indicators and predict three world residual coordinates. All use Adam 0.001, weight decay 0.0001, uniform 256-row minibatches and fixed final checkpoints. Direct/factual/nuisance models get 1000 updates; the compute control gets 3000 factual updates, matching direct plus two nuisance update counts but not asserting identical FLOPs or output capacity. All scales use fit rows only. A per-seed global mean of residualized FIT contrasts and zero effect are additional strong controls.

Fresh test risk uses original observed residuals and known propensities, without nuisance fitting on test. The gate requires a one-sided 95 percent upper risk bound below zero against every control and favorable direct-minus-factual point in every seed. Bootstrap preserves the same six paired strata as above.

| Direct minus control    | Point (mm <sup>2</sup> ) | Central95%      | Upper95% |
|-------------------------|--------------------------|-----------------|----------|
| factual                 | 23.977                   | [2.024,48.106]  | 44.201   |
| factual compute matched | 24.278                   | [2.370,48.434]  | 44.586   |
| global residualized     | 20.820                   | [-1.311,45.184] | 40.882   |
| zero                    | 20.541                   | [-1.223,45.436] | 40.638   |

Table7. Direct randomized learner minus each frozen control, mm squared; every fixed gate fails.

Direct-minus-factual seed points 32.35/19.35/20.23 all have the wrong direction. The direct method is UNPROMISING and ends without extra updates or label clipping. Fit/test take 30.31/134.49 seconds on separate admitted GPUs, producing 3.41/11.89 MiB. Six nuisance-fold scales, 16 fit model/data files, held predictions, all risk arrays and bootstrap intervals pass independent CPU/NumPy audits.

After retaining this primary failure, exploratory comparisons between the already frozen controls suggest factual-minus-zero/global differences of -3.436/-3.158 mm squared, with one-sided 95 percent upper bounds -1.106/-1.222. More factual updates have no robust incremental advantage over 1000 updates. These comparisons were selected after primary scoring; they are candidate-selection evidence, not a successful primary trial or independent Validation. A standard factual predictor is not a new method. Its candidate value must be tested in a fresh randomized closed-loop retained-success/drop experiment before any policy claim.

## 8. Limitations and research decision

These experiments involve one object, three shared motions, two existing actors and small fixed fitting procedures. Contact proxies plus sampled proximity are not attributed pairwise forces. Short-horizon measurement is not stable lift, retention or manipulation utility. No policy training, unseen-object/hand evaluation or hardware deployment is presented. Optimization seeds do not create independent physical datasets. Existing papers' claims are not replaced by these diagnostic tests.

The failed direct learner is stopped. The next decisive experiment freezes the simpler factual controller, compares native actor, random local corrections and state-independent fit-response selection, and measures full first-episode retained success and drops. A favorable first-step vertical displacement cannot replace retention benefit. No policy outcome has yet been collected for this design.

The original user objective remains a journal-level research outcome. This working paper does not meet it: a distinctive method and consequential matched task evidence remain necessary. Negative outcomes and the unresolved causal screen are retained to guide further work rather than converted into positive conclusions.

## Reproducibility

All new work is isolated in branch agent/contact-response-cm. Original checkpoints, motion assets and prior evidence are read-only. Physical/differential code is fixed at 466 e 80 d/44 f 22 ea; factorization at bbd 5 ba 4; fresh collection at 59 de 1 cf; null recovery at 33 d 1085. Manifests preserve commands, input SHA, GPU UUID, budgets and failures. Randomized collection uses 510 a 00 e; direct fitting 2075314; independent auditing f 283 bb 9. Every table is generated directly from recorded result JSON, including the failed fresh matching audit. This PDF is a ReportLab review copy; portable LaTeX source and tables are also retained. The first manuscript revision is preserved separately.

## References

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